

For the week of September 08-12

- 1) A tank is leaking fluid at a rate of $R(t)m^3/h$, where t is measured in hours.
- (a) Write a definite integral that calculates the amount of fluid lost in the first 3 hours.
- (b) If we know that $1 \leq R(t) \leq 3$ for $0 \leq t \leq 3$, give an upper and lower estimate of the amount of water lost in the first three hours.
- 2) Let $f(t)$ be the rate of change of earth's population, where t is measured in years.

- (a) Explain in words what is being calculated by the integral

$$\int_{2000}^{2025} f(t)dt.$$

- (b) The population of the earth was 6.1 billion people in the year 2000. Write an expression for the population of earth in 2025 using the function $f(t)$.

- 3) In the following, let $f(x) = F'(x)$. Write the integral $\int_a^b f(x)dx$ and evaluate it using the fundamental theorem of calculus.

(a) $F(x) = \sin(\pi x)$, with $a = -1/2$ and $b = 1/2$.

(b) $F(x) = \frac{4x}{1+x^2}$, with $a = -5$ and $b = 7$.

(c) $F(x) = \sqrt{-x}$, with $a = -9$ and $b = -4$.

- 4) Let $f(y)$ be the piecewise function defined by

$$f(y) = \begin{cases} -1 & y < 0 \\ y^2 & 0 \leq y \leq 1 \\ 1 & y = 1 \end{cases}$$

For each of the following, determine if we can evaluate the integral $\int_a^b f(y)dy$. If yes, evaluate it using the fundamental theorem of calculus.

(a) $\int_{-1}^2 f(y)dy$

(b) $\int_{-10}^{-1} f(y) dy$

(c) $\int_0^4 f(y) dy$

- 5) Use the fundamental theorem of calculus and theorems about the definite integral (§5.4 in textbook) to evaluate the following integrals.

(a) $\int_0^1 3x^2 + x - 5 dx$

(b) $\int_{-\pi^2}^{\pi^2} \sin(x) dx$

(c) $\int_{-\pi^2}^{\pi^2} |\sin(x)| dx$

(d) $\int_{-2025}^{2025} \sin(x^{71}) - 2x^7 dx$

(e) $\int_{10}^0 e^x dx$

- 6) Let $f(x) = \sin(x)$, $g(x) = 1 + 1/2x$ and $h(x) = xe^x$. If

$$A = \int_1^{100} h(x) dx \quad B = \int_1^{100} g(x) dx \quad C = \int_1^{100} f(x) dx,$$

then determine which of the following is true *without* evaluating any integrals.

(a) $C \leq A \leq B$

(b) $B \leq A \leq C$

(c) $C \leq B \leq A$

(d) $A \leq C \leq B$

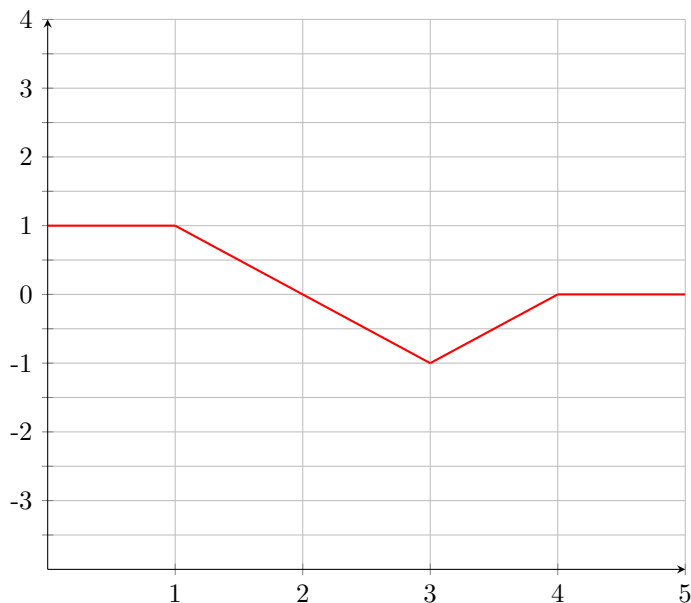
- 7) For each of the following functions $f(x)$, find the unique antiderivative $F(x)$ satisfying the additional assumptions.

(a) $f(x) = 3x^2 + 1$ with $F(0) = 1$

(b) $f(x) = \sin(x) + 4x$ with $F(\pi/2) = 6$

(c) $f(x) = \frac{1}{x}$ on the interval $[1, \infty)$ with $F(e) = 1$.

- 8) Let $f(t)$ be the function graphed below. If $F(t)$ is an antiderivative of $f(t)$, find the intervals on which $F(t)$ is increasing, decreasing, or constant. Do these intervals depend on which antiderivative of $f(t)$ we choose?



- 9) For $f(t)$ the same function as in problem 8, sketch the antiderivative $F(x)$ with
- (a) $F(0) = 0$
 - (b) $F(1) = 3$
 - (c) $F(2) = -3$

How are the graphs of the different antiderivatives related to one another?

- 10) Determine whether the following statements are true or false. If they are true, explain why. If they are false, explain why by giving a *counterexample*; i.e. give an example which shows the statement is not true.
- (a) If $F(x)$ and $G(x)$ are both antiderivatives of $f(x)$ then so too is $H(x) = F(x) + G(x)$.

- (b) If $F(x)$ is an antiderivative of $f(x)$ and $G(x)$ is an antiderivative of $g(x)$ then $F(x) + G(x)$ is an antiderivative of $f(x) + g(x)$.
- (c) If $f(x)$ is a positive function ($f(x) \geq 0$ for all x), then any antiderivative $F(x)$ is also a positive function.
- (d) If $f(x)$ is a continuous function with $\int_0^k f(x)dx = 0$ for every positive integer $k = 0, 1, 2, \dots$, and $F(x)$ is an antiderivative of $f(x)$, then $F(x)$ must be a constant function.

The following problems are for practice only and will not appear on quizzes.

- I) Textbook section 5.3, problems # 4-7, 16-19, 24-27, 34, 49, 50, 54
- II) Textbook section 5.4, problems # 42, 45-52, 57-60, 85-100
- III) Textbook section 6.1, problems # 1, 4, 5-12, 15, 18-21, 26, 30, 31, 37, 47